

OSSP

Practical- Assignment2

Name: Praneeth Kumar Gadila

Roll num: 2520030257

1.

```
GNU nano 8.7.1 practical2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int originalfile, copyfile;
    char buffer[100];
    int n;

    originalfile = open("textfile.txt",O_RDONLY);

    copyfile = open("textfile1.txt",O_WRONLY | O_CREAT , 0644);

    n = read(originalfile,buffer,sizeof(buffer));

    write(copyfile,buffer,n);

    close(originalfile);
    close(copyfile);

    printf("====the file has been copied succesfully====\n");

    return 0;
}

praneethgadila@praneethgadila:~$ nano practical2.c
praneethgadila@praneethgadila:~$ gcc practical2.c
praneethgadila@praneethgadila:~$ ./a.out
====the file has been copied succesfully=====
```

2.

```
praneethgadila@praneethgadila:~$ echo "now im having class in 15 lab , its o
s class" >sample_practical
praneethgadila@praneethgadila:~$ cat sample_practical
now im having class in 15 lab , its os class
praneethgadila@praneethgadila:~$ strace cat sample_practical
Command 'strace' not found, but can be installed with:
sudo apt install strace
```

```
praneethgadila@praneethgadila:~$ strace cat sample_practical
execve("/usr/bin/cat", ["cat", "sample_practical"], 0x7ffc9b8b8 /* 25 var
s */) = 0
brk(NULL)                               = 0x6489d42c8000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x708e976a0000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20761, ...}) = 0
mmap(NULL, 20761, PROT_READ, MAP_PRIVATE, 3, 0) = 0x708e9769a000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,
832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x708e97667
000
mmap(0x708e9766e000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x7000) = 0x708e9766e000
mmap(0x708e9768f000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x28000) = 0x708e9768f000
mmap(0x708e97696000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D
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openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,
      832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x708e97639000
mmap(0x708e9763d000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x708e9763d000
mmap(0x708e97661000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x708e97661000
mmap(0x708e97665000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x708e97665000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,
      832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x708e97513000
mmap(0x708e97525000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x708e97525000
mmap(0x708e975c5000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x708e975c5000
mmap(0x708e97637000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x708e97637000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"...,
      832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0"...,
        840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0

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8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OP0S
T|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXT
EN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x6489d430a000) = 0x6489d430a000
statx(AT_FDCWD, "sample_practical", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_m
ask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_siz
e=45, ...}) = 0
openat(AT_FDCWD, "sample_practical", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=45, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 45
splice(4, NULL, 1, NULL, 45, 0now im having class in 15 lab , its os class
) = 45
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x708e9769d000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
praneethgadila@praneethgadila:~$ cat trace.txt
cat: trace.txt: No such file or directory
praneethgadila@praneethgadila:~$ strace -o trace.txt cat sample_practical
now im having class in 15 lab , its os class
praneethgadila@praneethgadila:~$ cat trace.txt
execve("/usr/bin/cat", ["cat", "sample_practical"], 0x7ffc7c2af508 /* 25 var
s */) = 0
brk(NULL) = 0x568989758000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x732ab2446000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20761, ...}) = 0

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praneethgadila@praneethgadila:~$ cat trace.txt
execve("/usr/bin/cat", ["cat", "sample_practical"], 0x7ffc7c2af508 /* 25 var
s */) = 0
brk(NULL)                                = 0x568989758000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x732ab2446000
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20761, ...}) = 0
mmap(NULL, 20761, PROT_READ, MAP_PRIVATE, 3, 0) = 0x732ab2440000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,
832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x732ab240d
000
mmap(0x732ab2414000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x7000) = 0x732ab2414000
mmap(0x732ab2435000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x28000) = 0x732ab2435000
mmap(0x732ab243c000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D
ENYWRITE, 3, 0x2f000) = 0x732ab243c000
mmap(0x732ab243e000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_A
NONYMOUS, -1, 0) = 0x732ab243e000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEX
EC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,
832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x732ab23df
000
mmap(0x732ab23e3000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x4000) = 0x732ab23e3000
mmap(0x732ab2407000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x28000) = 0x732ab2407000
mmap(0x732ab240b000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D

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openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ..}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ..}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x732ab2045cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x732ab2045cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x732ab2045cb0}, {sa_handler=0x56897be730c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x732ab2045cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x732ab2045cb0}, {sa_handler=0x56897be730c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x732ab2045cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x56898979a000) = 0x56898979a000
statx(AT_FDCWD, "sample_practical", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=45, ..}) = 0
openat(AT_FDCWD, "sample_practical", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=45, ..}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ..}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 45
splice(4, NULL, 1, NULL, 45, 0) = 45
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0

```